
Predicting Student Dropout and Academic Success: A Supervised and Unsupervised Learning Study

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Abstract

This project analyzes student dropout, continued enrolment, and academic success in higher education using demographic, socioeconomic, administrative, and academic variables. Three tasks are addressed: multiclass classification of final student status, regression of second-semester average grade, and unsupervised learning through PCA and clustering. Across all tasks, first-semester academic performance emerges as the main source of information, while administrative and financial variables also play a relevant supporting role.

1 Introduction and data overview

Student dropout and academic success are major concerns for higher education institutions, both from an educational and from an organizational perspective. Anticipating adverse academic trajectories may help institutions identify at-risk students earlier and design targeted support strategies. In this context, machine learning provides a useful framework not only for prediction, but also for understanding which types of student information are most strongly associated with later outcomes.

The dataset used in this project contains information on students in higher education and includes variables from several complementary blocks: demographic and personal variables, access and prior academic context, socioeconomic and administrative variables, macroeconomic context, and academic performance during the first and second semesters. The original target variable defines three possible final student outcomes: *Dropout*, *Enrolled*, and *Graduate*. The original dataset variables and class labels are in Spanish; English translations are used throughout the report for readability. In addition, the dataset includes numerical academic-performance variables such as the number of curricular units enrolled, evaluated, approved, and the average grade obtained in each semester.

This project addresses three tasks. The first task is a multiclass classification problem aimed at predicting the final student status. The second task is a supervised regression problem in which the second-semester average grade is predicted from prior information only, carefully excluding variables that would introduce leakage. The third task is an unsupervised learning analysis designed to identify latent structures and student profiles in a reduced subset of interpretable numerical variables.

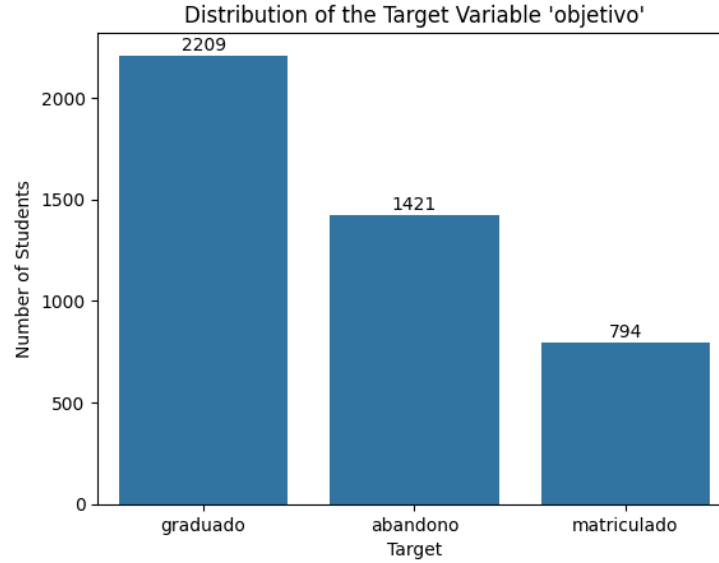


Figure 1: Distribution of the final student status variable.

1.1 Initial exploratory analysis

The exploratory analysis began by grouping variables into three broad types: continuous numerical variables, discrete numerical/count variables, and categorical variables. This distinction was useful both for descriptive analysis and for the later preprocessing pipelines. The dataset contains no missing values. Imputation was nevertheless kept inside the preprocessing pipeline for robustness and reproducibility, and the numerical and categorical preprocessing strategy was therefore designed conservatively, relying on imputation rather than row deletion.

Outliers were inspected using boxplots and the IQR criterion, especially for continuous academic variables. However, they were not removed automatically. In this problem, extreme observations may correspond to real and highly informative student profiles rather than to measurement errors, and removing them could distort the educational interpretation of the data. For that reason, a conservative data-preservation strategy was adopted unless there was clear evidence of inconsistency.

A particularly informative variable during the exploratory phase was the first-semester average grade, whose distribution already suggested that first-semester academic performance would likely play an important role in the later modeling stages.

1.2 Preprocessing and methodological decisions

Preprocessing decisions were driven by the structure of the three tasks. For supervised learning, train/test splits were created before fitting any preprocessing transformation. Numerical variables were imputed using the median and then standardized when appropriate, while categorical variables were imputed using the most frequent category and encoded through one-hot encoding. This was implemented through a common `ColumnTransformer`-based pipeline, which ensured that preprocessing remained consistent across models and avoided leakage from the test set.

For the classification task, two scenarios were defined. The *early* scenario only used variables available before observing university academic performance, while the *first-semester* scenario additionally incorporated first-semester academic variables. This comparison was designed to evaluate how much predictive signal is gained by including initial university performance.

For the regression task, the target variable was the second-semester average grade, `nota_media_2sem`. To keep the setup methodologically sound, all second-semester variables that were contemporaneous with, or directly derived from, second-semester academic performance were excluded. In particular, the excluded predictors were `asignaturas_2sem_convalidadas`, `asignaturas_2sem_matriculadas`,

asignaturas_2sem_evaluadas, asignaturas_2sem_aprobadas, and asignaturas_2sem_sin_evaluacion. These variables describe the same academic period as the target and would therefore introduce data leakage by making the prediction depend on information not available before or during the target semester. As a result, the model only used entry data, socioeconomic and administrative information, macroeconomic context, and first-semester performance.

For the unsupervised task, a smaller subset of interpretable numerical variables was selected, emphasizing age, admission-related variables, first-semester performance, and macroeconomic context. These variables were standardized before applying Principal Component Analysis (PCA) and clustering methods, since both techniques are strongly affected by scale.

2 Classification

2.1 Task setup and evaluation

The first supervised task predicts the final student status, which has three possible classes: *Dropout*, *Enrolled*, and *Graduate*. Since this is a multiclass problem with moderate class imbalance, accuracy alone is not sufficient. Therefore, the main metric used in this section is **macro-F1**, which gives equal importance to all classes and prevents the majority class from dominating the evaluation. This metric is complemented with **balanced accuracy** and standard **accuracy**.

Two prediction settings were considered. In the *early* scenario, only variables available before observing university academic performance were used. In the *first-semester* scenario, first-semester academic variables were added to the predictor set. This comparison is very relevant, since it quantifies how much additional predictive power is gained once early university performance becomes available.

Two model families were compared: a linear and interpretable baseline based on multinomial logistic regression, and a nonlinear tree-based model based on random forests. Both models were evaluated using an 80/20 train/test split and 5-fold cross-validation on the training data. Class weighting was incorporated in the classifiers to reduce the impact of class imbalance.

2.2 Main results

Table 1 summarizes the classification performance of the four evaluated models. The most important result is that the inclusion of first-semester academic variables leads to a substantial improvement over the early-prediction setting. This pattern is consistent across cross-validation and held-out test performance, and indicates that first-semester academic performance contains a large amount of information about the final academic outcome.

Among the evaluated models, the two best-performing approaches are the first-semester versions of logistic regression and random forest. Their performance is very similar. Random Forest achieves the highest cross-validated macro-F1, whereas Logistic Regression obtains the best test macro-F1 and the best test balanced accuracy. The difference in test macro-F1 between both models is marginal, approximately 0.004 points. Therefore, Logistic Regression is selected as the main reference model mainly because it remains considerably more interpretable while achieving essentially the same predictive performance.

A secondary but informative result is that, in the early scenario, logistic regression clearly outperforms random forest in macro-F1 and balanced accuracy. This suggests that, when only pre-entry information is available, the linear model provides a more stable and balanced decision boundary, whereas the random forest tends to favor the dominant classes more strongly.

2.3 Error analysis

Figure 2 shows the confusion matrix of the selected model, LogReg_1sem. The model performs best on the two extreme classes, *Dropout* and *Graduate*, while the intermediate class *Enrolled* is more difficult to identify. For the Enrolled class, only 84 out of 159 students are correctly classified, corresponding to a recall of approximately 52.8%. This confirms that the intermediate class is substantially harder to identify than the two more clearly defined outcomes. This is unsurprising, as *Enrolled* represents a more heterogeneous academic situation: students in this class have neither

Model	CV			Test		
	Acc	Bal Acc	Macro-F1	Acc	Bal Acc	Macro-F1
LogReg_1sem	0.7126	0.6787	0.6713	0.7096	0.6673	0.6622
RF_1sem	0.7415	0.6642	0.6718	0.7322	0.6510	0.6581
LogReg_early	0.5923	0.5731	0.5620	0.6056	0.5868	0.5740
RF_early	0.6397	0.5606	0.5632	0.6316	0.5580	0.5599

Table 1: Classification performance for the two prediction settings. Macro-F1 is the main metric.

clearly failed nor clearly completed their studies, and their profiles overlap more strongly with the other two groups.

This result is important for interpretation. The classifier is not simply learning a generic academic ranking; rather, it is able to distinguish especially well between students who are already following clearly favorable trajectories and those who are showing stronger signs of academic risk. By contrast, the intermediate class remains inherently less sharply defined.

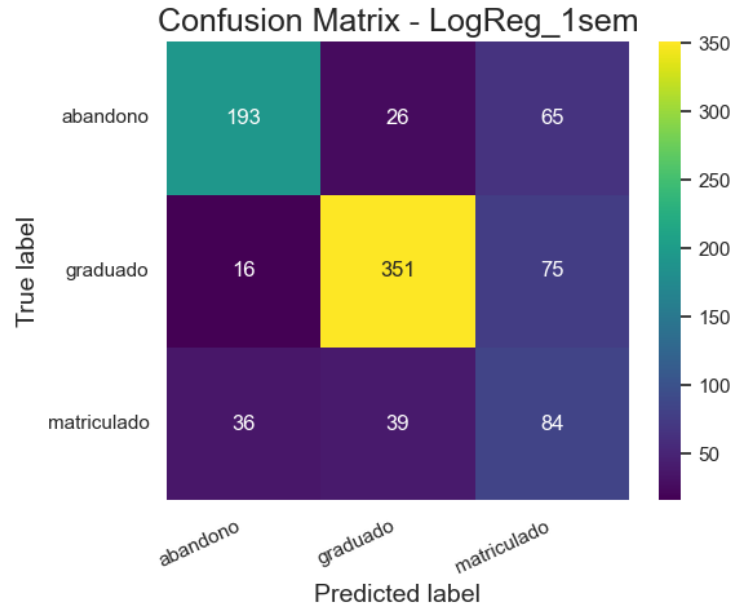


Figure 2: Confusion matrix of the selected classification model, LogReg_1sem.

2.4 Model interpretation

To interpret the selected classifier, the coefficients of the multinomial logistic regression model were examined after preprocessing. The most informative pattern is the strong role of first-semester academic performance. In particular, the number of curricular units approved in the first semester is strongly negatively associated with *Dropout* and strongly positively associated with *Graduate*. This is the clearest and most consistent predictive signal in the model.

Administrative and financial variables also play an important role. Being up to date with tuition payments is positively associated with *Graduate*, whereas payment difficulties are associated with *Dropout*. These effects are especially valuable from an institutional point of view, since they highlight that academic success is not only linked to academic performance, but also to administrative stability.

Overall, the interpretation of the multinomial logistic regression suggests that first-semester academic success and a stable administrative situation are strongly linked to favorable final outcomes. By contrast, the *Enrolled* class is less clearly characterized.

Class	Feature	Coefficient	Effect
Dropout	Tuition fees up to date = No	0.9420	positive
Dropout	Curricular units enrolled (1st semester)	0.7740	positive
Dropout	Tuition fees up to date = Yes	-0.8650	negative
Dropout	Curricular units approved (1st semester)	-1.7810	negative
Graduate	Curricular units approved (1st semester)	2.2070	positive
Graduate	Tuition fees up to date = Yes	0.6580	positive
Graduate	Tuition fees up to date = No	-0.6900	negative
Graduate	Curricular units enrolled (1st semester)	-1.0480	negative

Table 2: Selected coefficients of the LogReg_1sem model.

3 Regression

3.1 Task formulation

The second supervised task consists of predicting the student’s second-semester average grade. This task must be formulated carefully in order to avoid data leakage. In particular, second-semester variables that are contemporaneous with or nearly equivalent to the target cannot be used as predictors, since doing so would artificially inflate performance and invalidate the interpretation of the results. Therefore, the regression models in this section only use pre-entry variables, socioeconomic and administrative information, macroeconomic context, and first-semester academic performance. This formulation is especially relevant from an educational perspective. The goal is not merely to reconstruct the second-semester grade from redundant information, but rather to assess to what extent that outcome can be anticipated from information available earlier in the student’s trajectory.

3.2 Models and evaluation

Two regression models were compared. First, a **Ridge regression** model was used as a regularized linear baseline. This model is attractive because it is simple, stable, and interpretable, while also providing a strong linear baseline. Second, a **Random Forest Regressor** was used as a nonlinear alternative capable of capturing more complex interactions and nonlinear effects. Performance was evaluated using three complementary metrics: MAE, RMSE and R^2 . MAE provides an intuitive measure of the average absolute prediction error, RMSE penalizes large errors more strongly, and R^2 measures the proportion of variance explained by the model. As in the classification task, both cross-validation and held-out test performance were considered.

3.3 Main results

Table 3 summarizes the regression results. The overall picture is less clear-cut than in the classification task. The Random Forest regressor tends to perform better in cross-validation and achieves the best MAE, while Ridge remains highly competitive and performs slightly better on the held-out test set in RMSE and R^2 . This suggests that the nonlinear model may capture additional structure in the training data, but that the linear model remains a strong and stable baseline.

Given that Ridge achieves lower held-out test RMSE (2.644 vs. 2.708) and higher held-out test R^2 (0.742 vs. 0.729), it is selected as the main reference model for the regression task. However, this choice depends on prioritizing RMSE and R^2 : the Random Forest Regressor obtains a lower test MAE (1.397 vs. 1.593), indicating better average absolute error. For this reason, Random Forest is retained as a complementary nonlinear model and as an interpretability tool through its feature importances.

Model	CV			Test		
	MAE	RMSE	R^2	MAE	RMSE	R^2
Ridge	1.5723	2.6487	0.7389	1.5931	2.6441	0.7418
RF_Regressor	1.2729	2.4675	0.7726	1.3967	2.7076	0.7292

Table 3: Regression performance for Ridge and Random Forest Regressor.

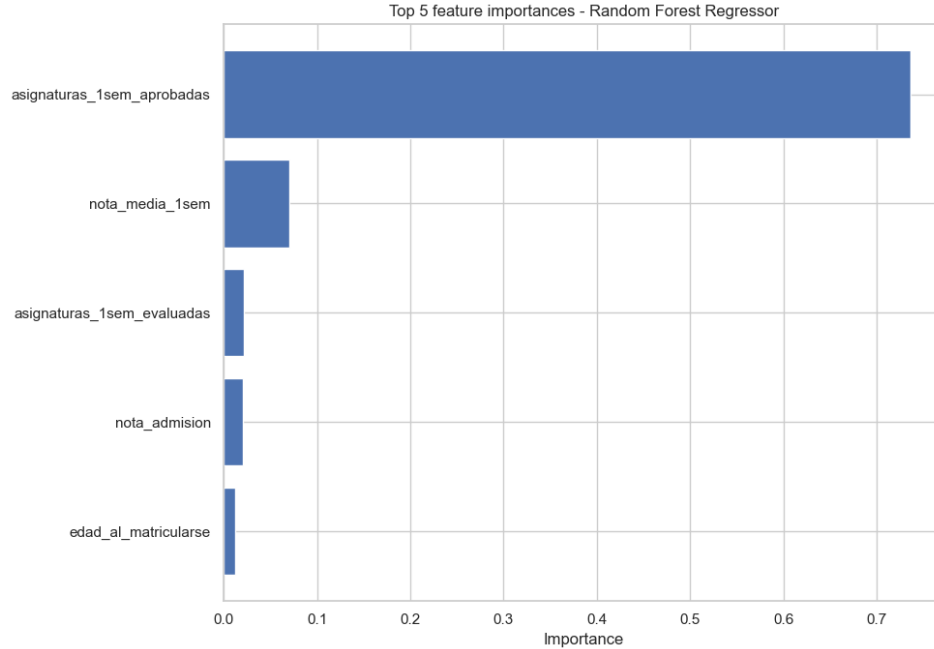


Figure 3: Top feature importances of the Random Forest regressor.

3.4 Model interpretation

To better understand which variables drive the regression predictions, the feature importances of the Random Forest regressor were examined. The clearest result is that first-semester academic performance is by far the strongest source of predictive information for the second-semester average grade. In particular, the number of curricular units approved in the first semester is the dominant predictor, with an importance score substantially larger than the rest of the variables.

Other first-semester variables, such as first-semester average grade and the number of evaluated curricular units, also contribute to the prediction, although to a much smaller extent. Admission grade and previous qualification grade retain some predictive value, whereas macroeconomic variables such as unemployment and inflation appear to play only a minor role.

Overall, these results suggest that early academic performance is much more informative of future academic outcomes than pre-entry or contextual variables. This conclusion is fully consistent with the classification results, where the inclusion of first-semester information also led to the largest increase in predictive performance.

4 Unsupervised learning

4.1 Motivation and methodological setup

The third task consists of applying unsupervised learning techniques to a reasoned subset of variables in order to identify latent student profiles and relevant structural patterns in the data.

The unsupervised analysis was structured around two complementary tools: **Principal Component Analysis (PCA)** and **clustering**. PCA was first applied to a standardized subset of interpretable numerical variables in order to obtain a lower-dimensional representation of the student profiles and to identify the main directions of variation in the data. Clustering was then used to search for groups of students with similar academic and contextual characteristics.

This setup is appropriate because it combines dimensionality reduction with group discovery, while keeping the analysis interpretable. In addition, both PCA and distance-based clustering are highly sensitive to scale, so the selected variables were standardized before the analysis.

4.2 Dimensionality reduction with PCA

The PCA results suggest that the dominant source of variation in the selected subset is strongly related to first-semester academic performance. In particular, variables such as the number of approved and evaluated curricular units in the first semester contribute prominently to the leading components. This

finding is highly consistent with the supervised tasks, where first-semester academic performance also emerged as the strongest source of predictive information.

The first two principal components explain 40.2% of the total variance in the selected standardized variables, with PC1 accounting for 25.1% and PC2 for 15.1%. Although this does not capture the full variability of the data, it provides a compact two-dimensional representation useful for visualization and interpretation.

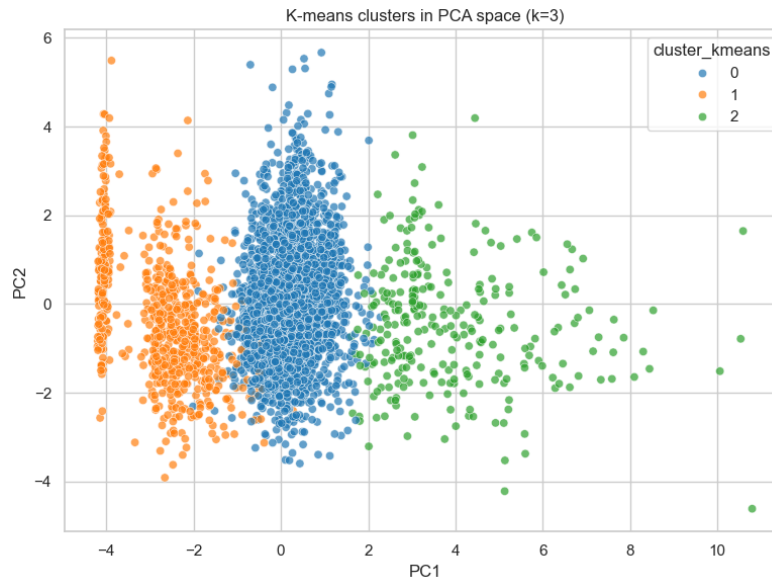


Figure 4: Two-dimensional PCA projection of the selected variables, colored by K-means cluster.

4.3 Clustering analysis

Clustering was performed using **K-means** as the main partitioning method. In addition, a **hierarchical clustering** analysis with Ward linkage was used as a complementary method, in order to assess whether a similar grouping structure could be observed from a different clustering perspective.

The number of clusters was selected through a combination of elbow and silhouette analysis. The silhouette score was highest for $k = 3$ among the main candidate solutions considered, with a value of 0.301, compared with 0.292 for $k = 2$ and 0.193 for $k = 4$. Although the absolute value indicates only moderate separation, the $k = 3$ solution provided the best balance between internal clustering quality and interpretability.

This solution shows a moderate but interpretable structure, mainly driven by first-semester academic performance. One cluster exhibits a markedly weaker academic profile, with lower numbers of approved and evaluated curricular units and worse overall first-semester performance. By contrast, the other two clusters correspond to stronger academic profiles, although with different levels of academic intensity and performance. This means that the clustering structure is not arbitrary: it captures meaningful differences in the students' early academic trajectories.

To further interpret the clusters, the original target variable was examined *a posteriori*. This step was used only for interpretation and not for cluster construction. The cluster with the weakest academic profile shows a substantially higher proportion of *Dropout* students, while the stronger clusters contain a larger proportion of *Graduate* students. This does not imply that the clusters reconstruct the target classes exactly, nor was that the goal. Rather, it shows that the unsupervised structure discovered from the selected variables is meaningfully related to later academic outcomes.

Cluster 1 deserves particular attention: 79.4% of the students assigned to this cluster are *Dropout* students, making it the most clearly at-risk group in the dataset. This cluster is characterized by markedly lower numbers of approved and evaluated curricular units in the first semester, as well as worse average first-semester grades — consistent with students who struggle academically from the very beginning of their university trajectory. From an institutional perspective, this profile is especially actionable: students falling into this cluster could be identified as early as the end of the first semester, making it a natural intervention point for academic support programs.

Cluster	Dropout	Graduate	Enrolled
0	0.229	0.571	0.201
1	0.794	0.107	0.099
2	0.237	0.638	0.125

Table 4: Interpretive summary of the K-means cluster profiles.

4.4 Discussion

The unsupervised results support the project’s main conclusion: student differences are largely driven by early academic performance, especially first-semester approved and evaluated curricular units. At the same time, the unsupervised task must be interpreted with appropriate caution. Unlike supervised learning, unsupervised methods do not offer a single standard evaluation criterion, so their assessment relies more heavily on interpretability, internal consistency, and visual inspection. For this reason, the value of this section lies less in claiming a unique “correct” partition of the data and more in showing that the dataset contains stable and educationally meaningful student profiles.

5 Conclusions and limitations

This project addressed three complementary machine learning tasks on a higher-education student dataset: multiclass classification of final academic status, regression of second-semester average grade, and unsupervised learning for profile discovery. Although the tasks were different, all three analyses pointed to the same central conclusion: **first-semester academic performance is the strongest source of information about later student outcomes.**

In the classification task, the inclusion of first-semester variables led to a clear improvement over the early-prediction setting. The best overall results were obtained by the first-semester versions of Logistic Regression and Random Forest, with very similar performance. Because multinomial Logistic Regression achieved slightly better held-out test performance while remaining substantially more interpretable, it was selected as the main reference classifier. Its coefficients showed that first-semester approved curricular units and tuition-payment status were among the most informative predictors of final student outcome.

In the regression task, the results again emphasized the importance of first-semester performance. Neither Ridge nor Random Forest dominated across all regression metrics, but both highlighted first-semester performance as much more informative than contextual variables. Similarly, PCA and clustering showed that the dataset is mainly structured around first-semester academic variables, with one cluster displaying a weaker trajectory and a higher proportion of dropout students.

In the unsupervised task, PCA and clustering provided an exploratory view of the internal structure of the dataset. The main axes of variation were again strongly linked to first-semester academic variables, and the clustering analysis revealed student groups with clearly different academic profiles. In particular, one of the clusters was characterized by a substantially weaker first-semester trajectory and also showed a much higher proportion of dropout students when compared *a posteriori* with the original target variable.

From an institutional perspective, these results suggest that student trajectories become significantly more predictable once first-semester performance is observed. They also indicate that administrative stability, particularly being up to date with tuition payments, is meaningfully associated with more favorable outcomes. Therefore, interventions aimed at improving early academic follow-up and reducing administrative or financial barriers may be especially relevant in practice.

At the same time, the conclusions of this study must be interpreted with caution. First, this is an observational dataset, and therefore the reported associations should not be interpreted causally. Second, some results—especially those involving one-hot encoded coefficients or cluster assignments—may depend on the exact sample and modeling choices. Third, unsupervised learning is inherently exploratory and does not provide a unique or universally validated structure of the data. Finally, while the models provide useful predictive and descriptive insights, they should be understood as decision-support tools rather than as definitive explanations of student success or dropout.

Overall, the project shows that a coherent combination of supervised and unsupervised learning can provide a meaningful picture of academic trajectories in higher education. More importantly, it suggests that the first semester plays a pivotal role in shaping later outcomes, making it a particularly valuable stage for monitoring and support.

References

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- [2] Realinho, V., Vieira Martins, M., Machado, J., and Baptista, L. (2021) *Predict Students' Dropout and Academic Success*. UCI Machine Learning Repository.